SHEET # 5

5.1 The following transfer functions represent three different filters meeting identical amplitude frequency response specifications:

$$H_1(Z) = \frac{a_o + a_1 Z^{-1} + a_2 Z^{-2}}{1 + b_1 Z^{-1} + b_2 Z^{-2}} \qquad H_2(Z) = \frac{a_3 + a_4 Z^{-1} + a_5 Z^{-2}}{1 + b_3 Z^{-1} + b_4 Z^{-2}}$$, and

$$H_3(Z) = \sum_{k=0}^{22} h_k Z^{-k}$$

Where:

$a_o = 3.136362 \times 10^{-1}$ $a_1 = -5.456657 \times 10^{-2}$ $a_2 = 4.635728 \times 10^{-1}$
$a_3 = -5.456657 \times 10^{-2}$ $a_4 = 3.136362 \times 10^{-1}$
$b_1 = -8.118702 \times 10^{-1}$ $b_2 = 3.339288 \times 10^{-1}$ $b_3 = -2.794577 \times 10^{-1}$
$b_4 = 3.030631 \times 10^{-1}$
and

$h_o = 0.39826480 \times 10^{-1} = h_{22}$
$h_1 = -0.16874380 \times 10^{-1} = h_{21}$
$h_2 = 0.34781130 \times 10^{-1} = h_{20}$
$h_3 = 0.12052890 \times 10^{-1} = h_{19}$
$h_4 = -0.44731860 \times 10^{-1} = h_{18}$
$h_5 = 0.27894610 \times 10^{-1} = h_{17}$
$h_6 = -0.87573360 \times 10^{-1} = h_{16}$
$h_7 = -0.90972060 \times 10^{-1} = h_{15}$
$h_8 = -0.15667550 \times 10^{-1} = h_{14}$
$h_9 = -0.28499560 \times 10^{0} = h_{13}$
$h_{10} = 0.74035030 \times 10^{-1} = h_{12}$
$h_{11} = 0.62349560 \times 10^{-10}$

(a) $H(Z) = H_1(Z) \times H_2(Z)$ (b) $H(Z) = H_1(Z) + H_2(Z)$ (c) $H(Z) = H_3(Z)$
(d) $H(Z) = [\, H_1(Z) \times H_2(Z) \,] + H_3(Z)$ (e) $H(Z) = H_1(Z) + H_2(Z) + H_3(Z)$

For each of the above filters find the following:

1- State whether it is an FIR or IIR filter or neither.

2- Represent the filter operation in a block diagram form an write down the difference equation.

3- Determine and comment on the computational and storage requirements.

5.2 The coefficients of Lattice filter are $K_1 = -0.266$ and $K_2 = 0.69$. Draw the realization diagram for the Lattice filter. Compute the impulse response coefficients for the filter and draw the equivalent transversal function.

5.3 A second-order IIR digital filter is characterized by the following transfer function:

$$H(Z) = \frac{1}{1 - 0.9Z^{-1} + 0.81Z^{-2}}$$

Draw the realization diagram for each of the following structure:

(a) Direct.

(b) Lattice.

Then, find the coefficients of the Lattice structure from the given transfer function.

5.4 The coefficients of transversal filter are $h_0 = h_1 = -0.266$ and $h_2 = 0.69$. Draw the realization structure for the transversal filter. Find the impulse response and the difference equation of the filter.

Assignment # 5

Give the realization structure of an IIR filter as shown in the following figure. Find the two-step difference equation relating the output, y (n), and input, x (n). Draw the equivalent Lattice filter's structure, and then compute its coefficients.

